

5 Courses

Programming for Everybody (Getting Started with Python)

Python Data Structures

Using Python to Access Web Data

Using Databases with Python

Capstone: Retrieving, Processing, and Visualizing Data with Python

UNIVERSITY OF MICHIGAN

Jun 16, 2025

Meixiang Cai

has successfully completed the online, non-credit Specialization

Python for Everybody

This Specialization builds on the success of the Python for Everybody course and will introduce fundamental programming concepts including data structures, networked application program interfaces, and databases, using the Python programming language. In the Capstone Project, you'll use the technologies learned throughout the Specialization to design and create your own applications for data retrieval, processing, and visualization.

Charles Severance
Clinical Associate Professor, School of Information
University of Michigan

The online specialization named in this certificate may draw on material from courses taught on-campus, but the included courses are not equivalent to on-campus courses. Participation in this online specialization does not constitute enrollment at this university. This certificate does not confer a University grade, course credit or degree, and it does not verify the identity of the learner.

Verify this certificate at:
https://coursera.org/verify/specialization/P2ZODTYXXS1Z



Imperial College London

3 Courses

Mathematics for Machine Learning: Linear Algebra

Mathematics for Machine Learning: Multivariate Calculus

Mathematics for Machine Learning: PCA

Jul 26, 2025



David Dye
Professor of Metallurgy
Department of Materials
Imperial College London

Samuel J. Cooper
Lecturer
Dyson School of Design Engineering
Imperial College London

Marc Deisenroth
Senior Lecturer
Department of Computing
Imperial College London

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has successfully completed the online, non-credit Specialization

Mathematics for Machine Learning

A sequence of 3 courses on the prerequisite mathematics for applications in data science and machine learning. Successful participants learn how to represent data in a linear algebra context and manipulate these objects mathematically. They are able to summarise properties of data sets and map them onto lower dimensional spaces with principal component analysis. Finally they can solve optimisation problems and use this skill to train models for describing data such as simple neural networks.

The online specialization named in this certificate may draw on material from courses taught on-campus, but the included courses are not equivalent to on-campus courses. Participation in this online specialization does not constitute enrollment at this university. This certificate does not confer a University grade, course credit or degree, and it does not verify the identity of the learner.

Verify this certificate at:
https://coursera.org/verify/specialization/Y41X42VCD72G

Universiteit Leiden

COURSE CERTIFICATE

Sep 10, 2025

Meixiang Cai

has successfully completed

Miracles of Human Language: An Introduction to Linguistics

an online non-credit course authorized by Universiteit Leiden and Meertens instituut (KNAW) and offered through Coursera

Professor Marc van Oostendorp, Dr.
Faculty of Humanities
Leiden University Centre for Linguistics

EDUCATION FOR EVERYONE · coursera · COURSE CERTIFICATE

Verify at:
https://coursera.org/verify/NB2K9UL06YOQ

Coursera has confirmed the identity of this individual and their participation in the course.

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百职帮

CERTIFICATE

Cai Meixiang

恭喜你完成了夜曲编程Python人工智能应用的学习
掌握了自然语言处理、图像识别、人脸识别、文字识别、
语音识别等多个知识点

学习课程

Python
人工智能应用

学习天数	运行代码行数	收获知识点
25	6968	69



LEARNING
RECORD

BAIZHIBANG BAIZHIBANG

Keep Coding

2025.05.11

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CERTIFICATE

Cai Meixiang

Congratulations on successfully completing the **Python Artificial Intelligence Applications** course at Yequ Programming.

You have mastered knowledge in **natural language processing, image recognition, face recognition, text recognition, and speech recognition**.

Course: Python Artificial Intelligence Applications

Study Days: 25

Lines of Code Executed: 6968

Knowledge Points Acquired: 69

Keep Coding

Date: 2025.05.11

I, Sun Zejing, translator of E-TRANSTAR (Beijing) Information Technology Co., Ltd., confirm this is a true and accurate translation of the original document.
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Signature: Sun Zejing Date of Translation: October 3, 2025

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百职帮

CERTIFICATE

Meixiang Cai

恭喜你完成夜曲编程Python网络爬虫课程的学习
掌握了网页请求和解析、文字信息爬取、图片信息爬取、反爬虫策略等应用知识，并完成6个实践项目。

学习课程

Python 网络爬虫

学习天数	运行代码行数	收获知识点
25	8129	65



LEARNING RECORD

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2025.03.01

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CERTIFICATE

Cai Meixiang

Congratulations on successfully completing **Python Web Crawling** course at Yequ Programming.

You have mastered knowledge and applications in **web requests and parsing, text information crawling, image information crawling, anti-crawling strategies**, and completed 6 practical projects.

Course: Python Web Crawling

Study Days: 25

Lines of Code Executed: 8129

Knowledge Points Acquired: 65

Keep Coding

Date: 2025.03.01



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CERTIFICATE

Cai Meixiang

恭喜你完成了夜曲编程Python数据分析·进阶的学习，掌握了**智能推荐、关联规则、回归分析**等多种算法的原理与模型的应用

学习课程

Python 数据分析·进阶

学习天数	运行代码行数	收获知识点
25	7036	61



LEARNING RECORD

Keep Coding

2025.09.05

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CERTIFICATE

Cai Meixiang

Congratulations on successfully completing the **Python Data Analysis · Advanced** course at Yequ Programming.

You have mastered the principles and applications of various algorithms and models, including **intelligent recommendation, association rules, and regression analysis**.

Course: Python Data Analysis · Advanced

Study Days: 25

Lines of Code Executed: 7036

Knowledge Points Acquired: 61

Keep Coding

Date: 2025.09.05

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百职帮

CERTIFICATE

Meixiang Cai

恭喜你完成了夜曲编程Python数据分析·基础的学习，掌握了**数据分析三大工具库、数据清洗、数据可视化**等多个知识点，并完成5个实践项目

学习课程

Python 数据分析·基础

学习天数	运行代码行数	收获知识点
25	4497	73



LEARNING RECORD

BAIZHIBANG

Keep Coding

2025.06.23

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CERTIFICATE

Cai Meixiang

Congratulations on successfully completing the **Python Data Analysis · Foundation** course at Yequ Programming.

You have mastered knowledge in the **three major libraries for data analysis, data cleaning, and data visualization**, and completed 5 practical projects.

Course: Python Data Analysis · Foundation

Study Days: 25

Lines of Code Executed: 4497

Knowledge Points Acquired: 73

Keep Coding

Date: 2025.06.23

I, Sun Zejing, translator of E-TRANSTAR (Beijing) Information Technology Co., Ltd., confirm this is a true and accurate translation of the original document.
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百职帮

CERTIFICATE

Cai Meixiang

恭喜你完成夜曲编程Python与SQL数据处理的学习，掌握了**基础查找、聚合统计、多表处理、Python交互**等多个知识点，并完成10+实践项目。

学习课程

Python与SQL数据处理

学习天数	运行代码行数	收获知识点
25	1100	61



LEARNING RECORD

BAIZHIBANG BAIZHIBANG

Keep Coding

2025.10.28

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CERTIFICATE

Cai Meixiang

Congratulations on successfully completing the **Python and SQL Data Processing** course at Yequ Programming.

You have mastered multiple skills including **basic queries, aggregate statistics, multi-table processing, and Python—SQL interaction**, and have completed more than 10 practical projects

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Course: Python and SQL Data Processing

Study Days: 25

Lines of Code Executed: 1100

Knowledge Points Acquired: 61

Keep Coding

Date: 2025.10.28

I, Niu Mengyuan, translator of E-TRANSTAR (Beijing) Information Technology Co., Ltd., confirm this is a true and accurate translation of the original document.
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Signature: Niu Mengyuan Date of Translation: October 28, 2025
翻译专用章 Valid for Translation Only

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Python入门课

CERTIFICATE

Cai Meixiang

恭喜你完成夜曲编程Python入门课的学习

掌握了**数据运算、条件判断、循环、函数、类与对象**等多个知识点

完成了2个实践项目

百职帮

学习天数	代码行数	收获知识点
31	1993	96



Keep Coding

2025.02.21

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CERTIFICATE

Cai Meixiang

Congratulations on successfully completing the **Python Introductory Course** at Yequ Programming.

You have mastered knowledge in **data operations, conditional statements, loops, functions, classes and objects**, and completed 2 practical projects.

Course: Python Introductory Course

Study Days: 31

Lines of Code Executed: 1993

Knowledge Points Acquired: 96

Keep Coding

Date: 2025.02.21

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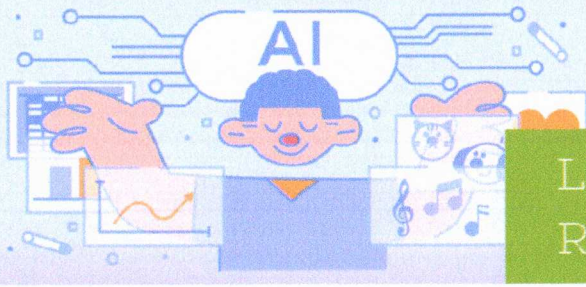
CERTIFICATE

Cai Meixiang

恭喜你完成夜曲编程AI大模型应用开发的学习，**掌握了AI大模型的工作原理、模型API的调用、LangChain和RAG开发等多个知识点**，并完成3个实战项目。

学习课程

AI大模型应用开发



LEARNING RECORD

学习天数	运行代码行数	收获知识点
15	7113	44

Keep Coding

2026.03.29

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CERTIFICATE

Cai Meixiang

Congratulations on successfully completing the **AI Large Model Application Development** course at Yequ Programming.

You have mastered knowledge in **the working principles of AI large models, model API calls, and LangChain and RAG development,** and completed **3 hands-on projects.**

Course: AI Large Model Application Development

Study Days: 15

Lines of Code Executed: 7113

Knowledge Points Acquired: 44

Keep Coding

Date: 2026.03.29

I confirm this is an accurate translation of the original document.
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The Date: October 25, 2026